

Chapter 2: Describing Data: Frequency Tables and Graphs

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Introduction

In this chapter, we are interested in the various ways in which we can present **qualitative** and **quantitative** data.

Qualitative Data

The Bahraini Minister of Urban Planning has received a complaint that petrol station queue times are very long. The minister asks you to collect data on all petrol stations in order to conduct a study.

Qualitative Data

After collecting 180 entries, your dataset is shown as below.

i	Petrol Station Name	Town	Governorate
1	Al Fateh Service Station	Manama	Capital
2	Bapco Energies Station – Muharraq	Muharraq	Muharraq
3	Bapco Energies Station – Sanabis	Sanabis	Capital
4	Salmabad Service Station	Salmabad	Northern
5	Galali Petrol Station	Galali	Muharraq
6	Al Buhair Petrol Station	Riffa	Southern
⋮			
180	Hajjyat Petrol Station 180	Riffa	Southern

Qualitative Data

There are various ways to present this data to the minister. One way is to construct a **frequency table**.

To do this, we need to create a set of groups (called classes) and count up all the observations that fall under that class.

Qualitative Data

Let us start with a basic analysis and find out how many petrol stations are available in each governorate:

Governorate	Frequency
Northern	52
Capital	40
Southern	45
Muharraq	43
Total	180

Qualitative Data

This is a simple way of presenting this information to the Minister. Notice that there are 4 governorates: Northern, Capital, Southern and Muharraq.

For the frequency table, we call each of these categories as **classes**. We call this a frequency table because it shows the total amount of **observations** for each **class**.

For example, there are 40 petrol stations in the Capital governorate.

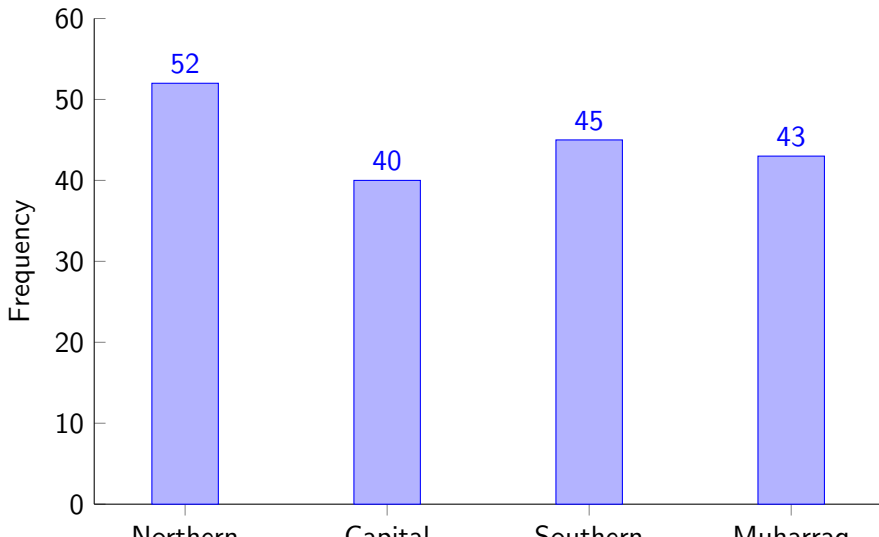
Qualitative Data: Bar Chart

You can take your dataset on petrol stations and display it in a **bar chart**.

This is when we have the frequency of the data on the vertical axis while each class is represented by a bar on the horizontal axis.

Qualitative Data: Bar Chart

Figure: Frequency of Petrol Stations by Governorate



Qualitative Data: Pie Chart

It is probably more informative to display this information with the **share** of each class to the overall total.

To do this, we simply divide the number of petrol stations from each class with the total. We call this the **relative frequency**.

$$\text{Relative Frequency} = \frac{\text{Frequency}}{\text{Total Frequency}}$$

Qualitative Data: Pie Chart

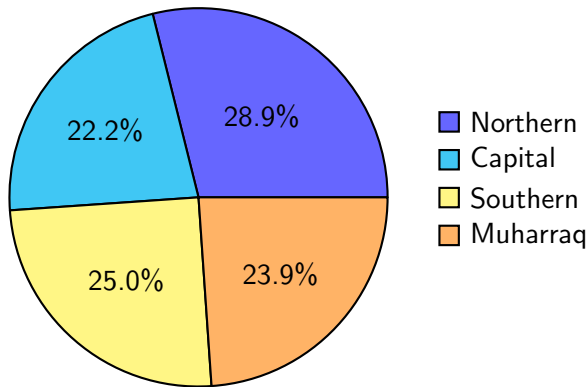
Since there are a total of 180 petrol stations, we can arrive at the following result:

Governorate	Frequency	Calculation	Relative Frequency
Northern	52	$\frac{52}{180}$	0.289
Capital	40	$\frac{40}{180}$	0.222
Southern	45	$\frac{45}{180}$	0.250
Muharraq	43	$\frac{43}{180}$	0.239
Total	180	$\frac{180}{180}$	1.000

Qualitative Data: Pie Chart

You can use this information to construct a **pie chart**. This displays the relative frequency of each class.

Figure: Petrol Stations by Governorate



Qualitative Data: Activity 1

Activity 1:

Bahrain City Center has the following aggregated information about types of stores in their mall:

Store Type	Frequency
Clothing and Fashion	70
Food and Beverage	45
Electronics	20
Health and Beauty	30
Home and Furniture	15
Entertainment and Leisure	20
Total	200

Qualitative Data: Activity 1

- (a) Create an additional column representing the relative frequency.
- (b) Display the totals of each class in a bar chart.
- (c) Display the relative frequency of each class in a pie chart.

Qualitative Data: Activity 1

Solution:

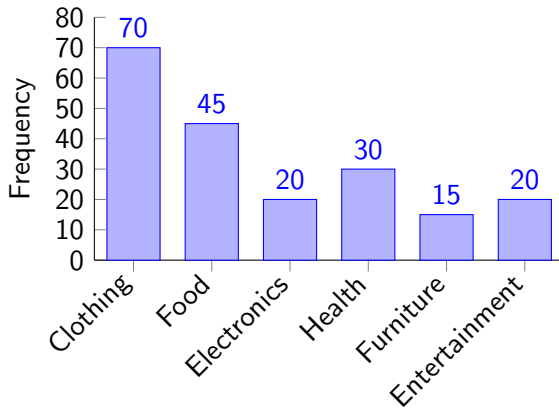
(a) To create the relative frequency column, we first get the total number of store in City Center. we then divide the frequency of each class with the total number of stores.

Store Type	Frequency	Relative Frequency
Clothing and Fashion	70	0.350
Food and Beverage	45	0.225
Electronics	20	0.100
Health and Beauty	30	0.150
Home and Furniture	15	0.075
Entertainment and Leisure	20	0.100
Total	200	1.000

Qualitative Data: Activity 1

(b) We can display the totals in a bar chart:

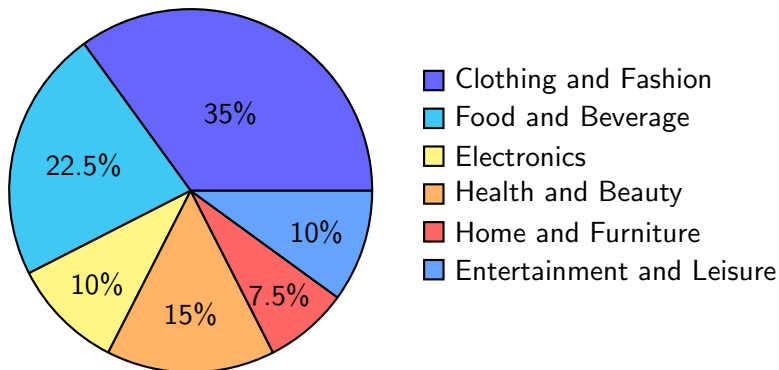
Figure: Frequency of Stores by Type



Qualitative Data: Activity 1

(c) We can display the relative frequency in a pie chart:

Figure: Relative Frequency of Stores by Type



Quantitative Data

Previously, we worked with displaying **qualitative** data as classes in a frequency table. This means that each class was some qualitative category, like the governorate or the type of store in a mall.

In this section, we will study a similar analysis, except this time the classes will be **quantitative** data.

Quantitative Data - Frequency Table

Consider a dataset on the sale of cars for the Toyota Showroom in September 2026:

Transaction Number	Sales Value (BD)
1	15,520
2	23,590
3	19,190
4	27,530
5	24,360
6	20,310
⋮	
180	30,380

Quantitative Data - Frequency Table

Sales data can have many observations. It would be useful to create a frequency table to properly summarise this information.

Just as before, we need to create a set of groups (classes). However, since we only have quantitative data, we would need to create the class based on an **interval of values**.

Quantitative Data - Frequency Table

Suppose we created the frequency of transactions under these classes:

- 2,000 up to 12,000
- 12,000 up to 24,000
- 24,000 up to 34,000

That is, there are three classes, and each one has a width of 10,000 BD.

Quantitative Data - Frequency Table

Interval Class (BD)	Frequency
2,000 up to 12,000	5
12,000 up to 24,000	75
24,000 up to 34,000	100
Total	180

Quantitative Data - Frequency Table

This is how we we would display a frequency table with quantitative data classes.

There are 5 transactions that are valued in the first class, 75 transactions in the second class, and 100 transactions in the third class.

Quantitative Data - Frequency Table

We arbitrarily selected very few classes to represent the dataset. Depending on the context of your study, you can also have many classes.

For example, lets instead have 14 classes, where each interval has a width of 1500 BD:

Quantitative Data - Frequency Table

Interval Class (BD)	Frequency
2,000 up to 4,500	1
4,500 up to 7,000	0
7,000 up to 9,500	1
9,500 up to 12,000	3
12,000 up to 14,500	5
14,500 up to 17,000	10
17,000 up to 19,500	13
19,500 up to 22,000	22
22,000 up to 24,500	35
24,500 up to 27,000	24
27,000 up to 29,500	31
29,500 up to 32,000	29
32,000 up to 34,500	6
34,500 up to 37,000	0
Total	180

Quantitative Data - Optimal Distribution

You should know that when deciding on the number of classes and the choice of interval width, we can follow a rule of thumb to construct the optimal amounts.

Quantitative Data - Optimal Distribution

You need to decide on **the number of classes** and the **width of the interval** for each class. In order to do this, you need three details about your dataset:

- The total number of observations n .
- The minimum value of your variable in the dataset min.
- The maximum value of your variable in the dataset max.

Quantitative Data - Optimal Distribution

The Number of Classes

Use the following rule:

$$2^k > n$$

where k is the number of classes and n is the total observations in your dataset.

Keep choosing different values of k until the equality is satisfied.

Quantitative Data - Optimal Distribution

Width of the Interval

To determine the optimal interval, we use the following rule:

$$i = \frac{\max - \min}{k}$$

max and min are the largest and smallest values of the variable in your dataset. We normally round i to the nearest whole number or a convenient multiple of 10, 100, 1000, etc.

Quantitative Data - Activity 2

Activity 2: Consider the sales data regarding Toyota's showroom. We are given the following details:

- $n = 180$
- $\min = 2,940$ BD
- $\max = 32,920$ BD

Calculate the optimal number of classes k and interval i using the rules described earlier.

Quantitative Data - Activity 2

Solution:

For the number of classes, we need to satisfy the rule:

$$2^k > n$$

Since there are $n = 180$ observations, we can test for values of k until the equality is just satisfied.

Quantitative Data - Activity 2

For example, if $k = 7$:

$$2^7 = 128 < 180$$

which does not satisfy the condition. However if we do $k = 8$:

$$2^8 = 256 > 180$$

Therefore, $k = 8$ is the recommended number of classes.

Quantitative Data - Activity 2

As for the interval width, we need to simply compute:

$$i = \frac{\text{max} - \text{min}}{k}$$

$$i = \frac{39290 - 2940}{8} = 4543.75$$

We can round this to the nearest whole number to get to 4,500.

Quantitative Data - Activity 2

Now that we know how to calculate the optimal number of classes and interval width, we can display this information in a frequency table:

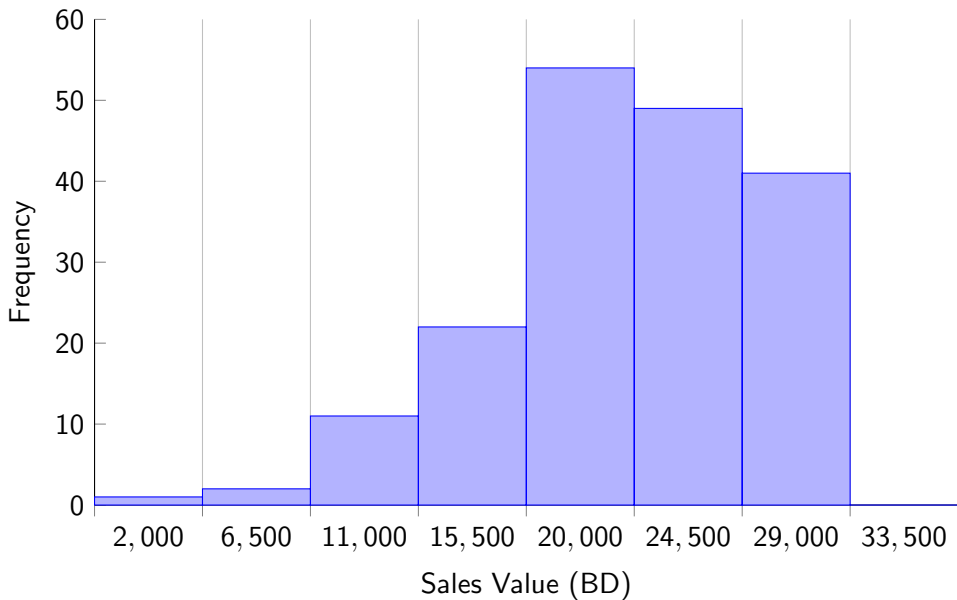
Interval Class (BD)	Frequency
2,000 up to 6,500	1
6,500 up to 11,000	2
11,000 up to 15,500	11
15,500 up to 20,000	22
20,000 up to 24,500	54
24,500 up to 29,000	49
29,000 up to 33,500	41
33,500 up to 38,000	0
Total	180

Quantitative Data - Histogram

We can take our optimised frequency table and display it in a **histogram**.

This is very similar to a **bar chart**, where the vertical axis is the frequency and the horizontal axis represents the class intervals:

Quantitative Data - Histogram



Quantitative Data - Frequency Polygon

Another way to display quantitative data is through a **frequency polygon**.

This is when the vertical axis has the frequency, but the horizontal axis is represented by the **midpoint** of every interval.

Quantitative Data - Frequency Polygon

To calculate the midpoint, simply apply the below formula for each interval class:

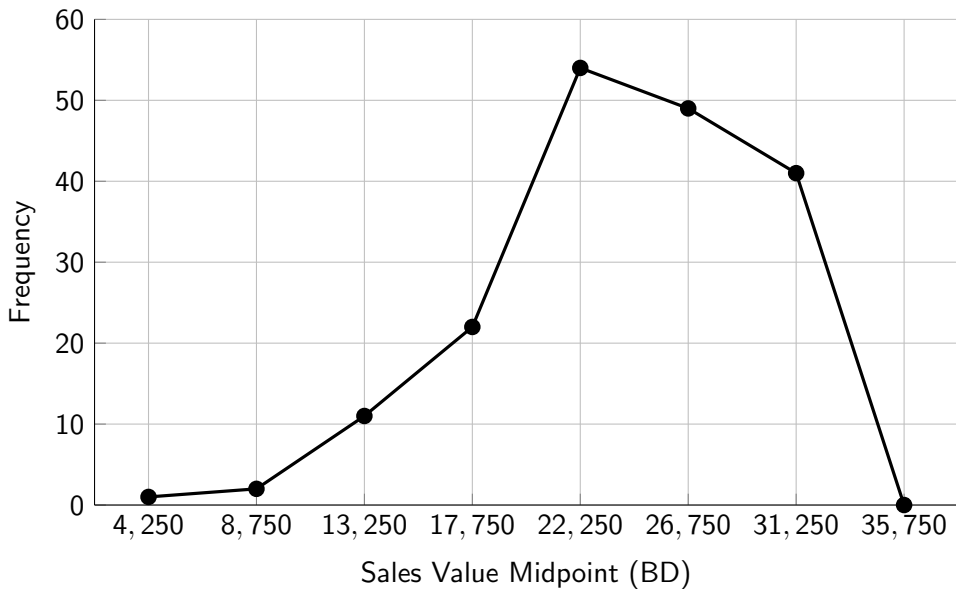
$$\frac{\text{Lower Bound} + \text{Upper Bound}}{2}$$

Quantitative Data - Frequency Polygon

Using the car sale data:

Interval Class (BD)	Midpoint (BD)	Frequency
2,000 up to 6,500	4,250	1
6,500 up to 11,000	8,750	2
11,000 up to 15,500	13,250	11
15,500 up to 20,000	17,750	22
20,000 up to 24,500	22,250	54
24,500 up to 29,000	26,750	49
29,000 up to 33,500	31,250	41
33,500 up to 38,000	35,750	0

Quantitative Data - Frequency Polygon



Quantitative Data - Relative Frequency

Similar to the previous section, we can also construct relative frequency distributions. Remember that the way to create this is take the total count of each class and divide it by the total number across all classes:

$$\text{Relative Frequency} = \frac{\text{Frequency}}{\text{Total Frequency}}$$

Quantitative Data - Relative Frequency

Interval Class (BD)	Frequency	Calculation	Relative Frequency
2,000 up to 6,500	1	$\frac{1}{180}$	0.006
6,500 up to 11,000	2	$\frac{2}{180}$	0.011
11,000 up to 15,500	11	$\frac{11}{180}$	0.061
15,500 up to 20,000	22	$\frac{22}{180}$	0.122
20,000 up to 24,500	54	$\frac{54}{180}$	0.300
24,500 up to 29,000	49	$\frac{49}{180}$	0.272
29,000 up to 33,500	41	$\frac{41}{180}$	0.228
33,500 up to 38,000	0	$\frac{0}{180}$	0.000
Total	180	$\frac{180}{180}$	1.000

Quantitative Data - Cumulative Frequency

A cumulative frequency graph is one way of illustrating quantitative data across classes.

Consider again the frequency graph for Toyota car sales:

Quantitative Data - Cumulative Frequency

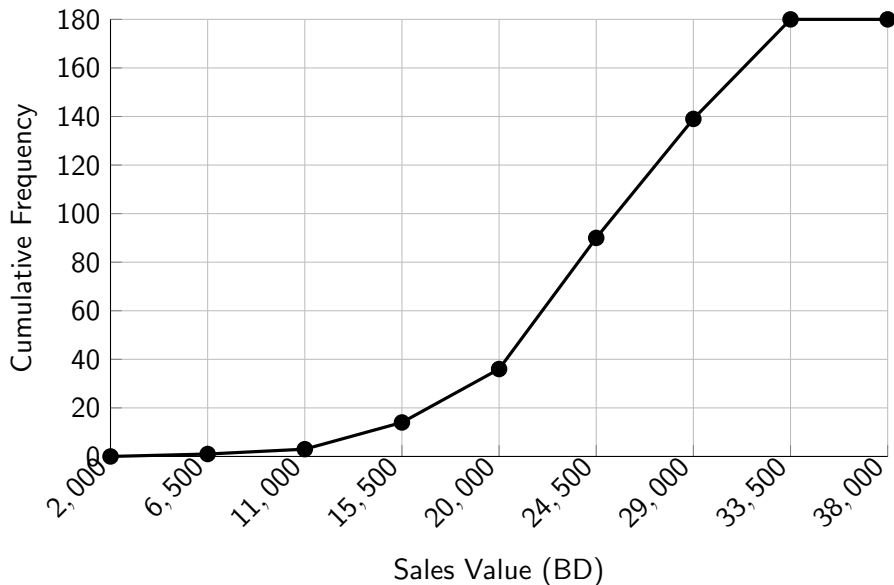
Interval Class (BD)	Frequency
2,000 up to 6,500	1
6,500 up to 11,000	2
11,000 up to 15,500	11
15,500 up to 20,000	22
20,000 up to 24,500	54
24,500 up to 29,000	49
29,000 up to 33,500	41
33,500 up to 38,000	0
Total	180

Quantitative Data - Cumulative Frequency

The cumulative frequency is simply the addition of each row:

Interval Class (BD)	Frequency	Calculation	Cumulative Frequency
> 6,500	1	1	1
> 11,000	2	$1 + 2$	3
> 15,500	11	$3 + 11$	14
> 20,000	22	$14 + 22$	36
> 24,500	54	$36 + 54$	90
> 29,000	49	$90 + 49$	139
> 33,500	41	$139 + 41$	180

Quantitative Data - Cumulative Frequency



Quantitative Data - Cumulative Frequency

You can also gather the cumulative relative frequency to assess the contribution of each class.

$$\text{Relative Cumulative Frequency} = \frac{\text{Cumulative Frequency}}{\text{Total Frequency}}$$

Quantitative Data - Cumulative Frequency

Interval (BD)	Cumulative Frequency	Calculation	Relative Cumulative Freq.
< 6,500	1	$\frac{1}{180}$	0.006
< 11,000	3	$\frac{3}{180}$	0.017
< 15,500	14	$\frac{14}{180}$	0.078
< 20,000	36	$\frac{36}{180}$	0.200
< 24,500	90	$\frac{90}{180}$	0.500
< 29,000	139	$\frac{139}{180}$	0.772
< 33,500	180	$\frac{180}{180}$	1.000

Quantitative Data - Cumulative Frequency

An illustration in which we have the cumulative frequency and the relative cumulative frequency is called an **Ogive**.

Quantitative Data - Cumulative Frequency

